Forecasting Regional Dengue Cases in The Philippines Using Historical Temporal and Geographical Data

Eunice Joanne G. Macasa
Mapua *Malayan Colleges Mindanao Undergrad School*
Davao City, Philippines
ejMacasa@mcm.edu.ph

Grant Ivan L. Sartiga
Mapua Malayan Colleges Mindanao Undergrad School
Davao City, Philippines
giSartiga@mcm.edu.ph

Tyra Denise L. Ventura
Mapua Malayan Colleges Mindanao Undergrad School
Davao City, Philippines
tdVentura@mcm.edu.ph

***Abstract* —** Dengue fever remains a significant public health concern in the Philippines, with seasonal outbreaks that vary across regions. Despite its widespread impact, there is a lack of forecasting tools that provide region-specific predictions to support timely and preventive health measures. This study addresses that gap by developing and applying a datadriven model to forecast monthly dengue cases across the country's 17 regions from 2026 to 2030. Using historical data from 2016 to 2020 obtained from Kaggle with records of monthly dengue cases and deaths, the study employed Prophet, a time-series forecasting tool developed by Facebook. Prophet is suited for capturing seasonal patterns, trends, and the influence of external factors. The dataset was cleaned, encoded, and organized by seasonal periods to ensure consistency across analytical tools such as JASP, Excel, and Python. The study objectives include identifying temporal and regional trends, forecasting future caseloads, and supporting public health planning. Results reveal that dengue transmission follows a seasonal pattern, with peaks typically occurring from July to September. Region IV-A recorded the highest historical case count, while BARMM had the lowest. The model's forecasts aim to strengthen early warning systems and guide regionspecific outbreak preparedness across the Philippines.

***Keywords* —** *Dengue forecasting, time-series analysis, Prophet model, regional health, seasonal patterns, Philippines, vector-borne disease, public health planning*

INTRODUCTION

Each year, thousands of Filipinos, especially children, are hospitalized due to dengue fever, a mosquito-borne disease that remains one of the most serious public health threats in the Philippines. According to the World Health Organization [1], the dengue virus is transmitted to humans through the bite of an infected female mosquito, primarily the Aedes aegypti species. Mosquitoes can become infected by viremic individuals who are symptomatic, presymptomatic, or asymptomatic. While the primary mode of transmission is through mosquito vectors, less common forms such as blood products, organ donation, transfusion, and transovarial spread in mosquitoes have also been documented. Most infected individuals experience no symptoms or only mild ones, such as fever, muscle and joint pain, headache, and nausea. However, severe disease can develop in rare cases involving plasma leakage, bleeding, and organ impairment [2]. According to WHO, dengue is now the most common and quickly spreading vectorborne illness worldwide [3], with 1.3 billion of the 3.5 billion people at risk residing in dengue-endemic areas across 10 SEA Region countries. In the Philippines, cases have increased by 6% in recent weeks, with 6,720 cases reported, and 52,008 cases were recorded between January 01 and February 22, 2025, 64% higher than the same period in 2024 [4]. Transmission is significantly influenced by environmental factors like temperature and rainfall, which affect mosquito breeding, along with geographical factors such as urban vs. rural and coastal vs. inland settings. According to WHO, urbanization and population movement have also contributed to the spread by supporting mosquito proliferation and facilitating virus

transmission [5]. Socioeconomic and geographic factors such as population size, access to clean water and sanitation, and urban development further shape dengue transmission patterns [6]. While dengue has historically been seen as an urban disease, a recent study in the Philippines challenges this view, showing that some rural areas may now face higher infection risks than urban areas [7].

Understanding the dengue transmission patterns and methodologies for forecasting dengue outbreaks is significant for effective public health management. This section explores epidemiological findings on dengue; it examines how environmental and socioeconomic factors contribute to the dengue outbreak and highlights the role of time series forecasting in public health. Dengue fever continuously to be recognized globally as a significant public health threat, along with its impact on scientific studies. The World Health Organization (WHO) highlights dengue as a rapidly spreading mosquitoborne disease, placing a large population at risk of dengue [1].

In the Philippines, dengue remains a significant threat to public health; this mosquito-borne disease contributes to thousands of hospitalizations annually [4]. Additionally, the influence of environmental and socioeconomic factors on dengue patterns has a significant impact on scientific studies. Numerous studies highlight the climate conditions, particularly during humidity and rainfall, which affect the mosquito's breeding, survival rates, and the replication cycle of the mosquito [10]. A study that has examined 61 provinces found that the relationship between temperature and dengue incidence varies across regions and is influenced by socioeconomic factors.

Additionally, this highlights that economic status, population size, and access to clean water and sanitation significantly affect how the disease spreads. Research by Snyder et al. [5] has implemented forecasting tools and visual aids for public health, decreasing the risks of dengue transmission.

Time series forecasting is a significant tool in epidemiology, widely used to predict outbreaks of infectious diseases such as influenza, malaria, and dengue. With these findings, this study addresses the significant gap in providing region-specific and seasonal forecasts for dengue cases in the Philippines. Through the combing of temporal and geographical data within a time series using a reliable forecast model, this study aims to create insights that can improve early warning systems and enhance public health preparedness for future dengue outbreaks in the Philippines.

This research addresses the gap in existing studies by focusing on forecasting dengue cases across different regions of the Philippines using both temporal and geographical data. Despite the rising number of dengue cases in the Philippines, there is still a lack of forecasting analysis on how climate and geographic patterns affect the spread of the disease over time. While the government collects data on dengue cases and deaths, this information is not consistently used to identify long-term trends or predict future outbreaks. This gap limits the ability of health authorities to prepare for and prevent future outbreaks, especially at the regional level, where environmental and socioeconomic conditions vary across the country.

This study aims to forecast dengue cases in the Philippines over five years, from 2026 to 2030, using the available regional and geographic data. The objectives of this study are to

1. To develop and implement a data science model, utilizing Prophet a time-series tool, capable of forecasting dengue cases around the Philippines based on historical, temporal, and geographical factors to distinct seasonal patterns and geographical variations in dengue outbreak.
2. To extract insights from forecasted data by identifying and analyzing significant trends and reoccurring patterns in monthly and regional cases.

This study is significant because it addresses a gap by focusing on forecasting dengue cases across different regions of the Philippines using both temporal and geographical data. It aims to improve early warning systems and strengthen public health preparedness by generating accurate and timely outbreak predictions for the years 2026–2030. The results of this study will help health officials in planning and response efforts. This data-driven approach plays a vital role in reducing the national and global impact of dengue, lowering illness and death

rates, and protecting the health and economic stability of communities across the country.

This study focuses on forecasting dengue cases across all 17 regions of the Philippines over five years, from 2026 to 2030. The forecasts are based on historical data from 2016 to 2020, which includes the monthly records of dengue cases and deaths. The study uses regional and temporal data to seek seasonal patterns and geographic variations in dengue transmission. The model aims to identify the peak periods and trends in dengue outbreaks to help improve regional public health planning.

The study is limited to the use of historical dengue cases from the years of 2016 to 2020. As a result, any significant dengue outbreaks that occurred before 2016 or after 2020 are not reflected in the data. While the analytical approach forecasts the dengue cases from 2026 to 2030, the forecast relies on the patterns identified in the historical data from 2016 to 2020. As a result, the accuracy may be limited, especially since the future environment or social conditions may differ significantly during the forecast period.

MATERIALS AND METHODS

A. Data Source

This study utilizes a single dataset obtained from Kaggle [20], an open-source data platform commonly used for machine learning and analytics projects. It contains historical records of dengue cases across various regions in the Philippines. Monthly observations structure the data set and include necessary information for temporal and regional analysis of dengue outbreaks, which is a valuable asset that directly supports this study's goal of forecasting regional dengue cases. The dataset provides valuable input for forecasting and modelling and was made publicly available with acknowledgement to the Department of Health of the Philippines for granting open access to the data.

The data we used includes monthly records of dengue cases and deaths from 2016 to 2020 across all 17 regions in the Philippines. Each entry has information on the month, year, region, and the number of cases and deaths. In total, there are 1,020 entries. The detailed regional data breakdown supports in-depth analysis and enables the development of region-specific health strategies. The data was cleaned and checked for consistency before being used in forecasting models. The chosen timeframe captures recent trends while providing enough historical data to build an effective model.

B. Data Reprocessing

The dataset underwent several steps to ensure it was ready for analysis. The dataset was checked for duplicates and incomplete entries; however, none were found, so no entries were corrected or removed. The data was maintained in CSV format throughout the process for compatibility across the tools. Column headers and date formats were ensured in both works in Excel, Jasp and Python. Since we were interested in seasonality, we had to turn the month names into seasonal groups that match how the climate usually works here. For example, months like July or August were labelled under the wet season. This recording helped us prepare the dataset for analysis. Everything was cleaned up and appropriately encoded so we could use it in JASP, Python and Excel.

C. Analytical Approach

In this study, we analyzed data to forecast dengue cases in the Philippines for the next five years (2026 to 2030) by using historical data containing temporal and regional information. We used a forecasting method called Prophet, which is designed to handle time series data with strong seasonal effects and historical trends. Developed by Facebook, Prophet is capable of modelling seasonality, holidays, and the impact of external factors. Its main objective is to forecast future values by capturing patterns in past data, including seasonal changes and other influential variables.

D. Conceptual Framework

This study uses historical dengue data to generate monthly forecasts for each region in the Philippines. The input variables include date, case count, and region. After the data was cleaned and formatted, it was passed into the Prophet forecasting model. Prophet processed the dataset by identifying recurring patterns over time. It automatically detected seasonality by using the time variable and modeled changes in case levels that repeat during certain months. The model was run separately for each region,

allowing it to learn regional patterns based on past trends. The output is a set of monthly case forecasts from 2026 to 2030, reflecting the patterns Prophet found in the earlier years. This framework links structured time-based data to a model that identifies and projects seasonal and trend-based shifts in dengue cases.

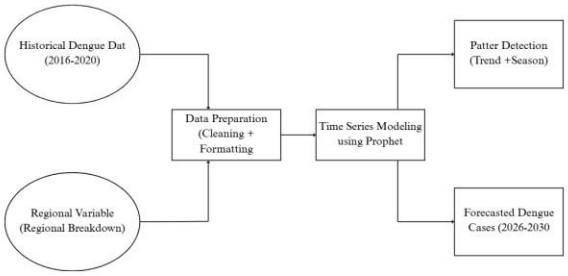


Fig.1 Conceptual Framework

RESULTS AND DISCUSSION

The results of this study reveal patterns and trends in dengue cases across the Philippines from 2016 to 2020 and provide forecasted values for 2026 to 2030 based on historical temporal and regional data.

Descriptive Analysis of Historical Data

Before applying forecasting models, the historical data from 2016 to 2020 was examined to identify trends and patterns.

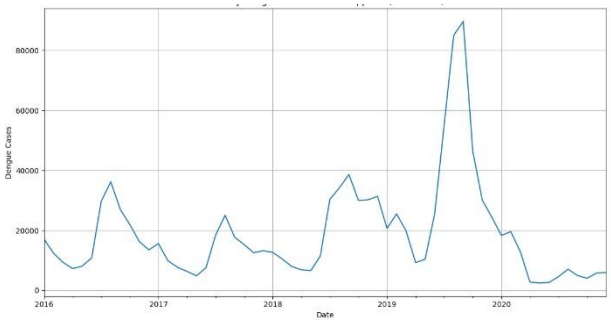


Fig.2 (Total Reported Dengue Cases in the Philippines (2016-2020))

To begin with, the Total number of reported dengue cases in the Philippines peaked in 2019, recording the highest number of cases during the 5 years with over 400,000 cases. The lowest year was observed during 2017, which had fewer cases than 150,000 cases.

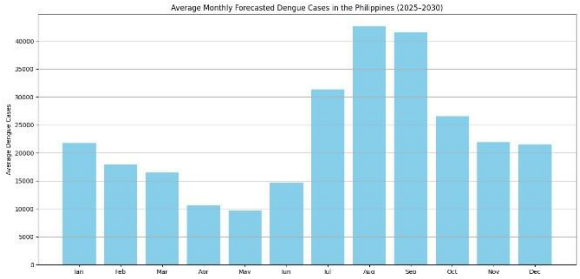


Fig.3 (Average dengue cases per month)

In terms of seasonality, Dengue cases in the Philippines show a consistent seasonal pattern. Lowest cases were observed during the dry season (March to May), with April recorded the lowest average of cases ,between 300 to 400 cases. A sharp increase in dengue cases starts in June, with cases peaking during July and August, reaching 2,000 to 2,500 cases. Peak months, July to September, show the highest number of cases corresponding to the external factors like rainfall season. This supports that the wetter conditions create favorable breeding grounds for mosquitoes. After September, there is a gradual decline in cases until December.

Fig.4 (Total cases per region)

Although these patterns represent national averages, different regions might exhibit distinct patterns due to the differences in environmental and geographic conditions. The region with the highest



total number of cases from 2016 to 2020 is Region IV-

A, with 160,000 cases. On the other hand, the region with the lowest total number of cases is BARMM (Bangsamoro Autonomous Region in Muslim
Mindanao), with a fever of 20,000 cases.

A. Forecasted Trends

a. Model Used

Utilizing historical data, the Prophet forecasting model was implemented to predict regional dengue cases in the Philippines for the years 2026 to 2030. The Prophet model was utilized for this study due to its ability to handle both non-seasonal and recurring seasonal patterns in time series data. It can also incorporate external factors, making it suitable for forecasting diseases like dengue that are influenced by seasonal changes and other environmental conditions.

b. Forecast Results by Region

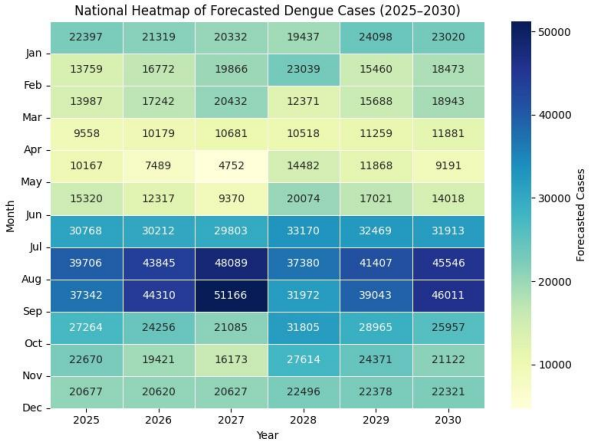


Fig.5 (Forecasted Dengue Cases per Region (20262030))

Forecasted dengue cases across the regions show varying trends but consistently peak during the rainy seasons each year. In BARMM, cases gradually increase over the years, possibly reaching 600 cases annually, peaking after mid-year. CAR shows a gradual decrease, with peaks around mid-year and starting from 500 cases. NCR maintains consistently low minimum cases but peaks after mid-year in 2027, predicting over 6,000 cases. Region I forecasts a gradual increase, with peaks surpassing 3,000 cases during rainy seasons. Region II also shows increasing trends, with yearly peaks exceeding 2,000 cases. In Region III, low cases remain steady, but 2027 may see peaks of 6,000 cases. Region IV-A presents increasing trends, with the highest peak reaching 12,500 in 2027. Region IV-B forecasts gradual increases, with yearly peaks surpassing 2,000. Region V shows a similar upward trend, with peaks reaching above 1,500. Region VI predicts steadily low cases, with 2027 peaks exceeding 8,000. In Region VII, average cases gradually decline, with early peaks of 2,000. Region VIII forecasts a 2027 peak of 5,000, with low cases gradually increasing yearly. Region IX shows gradual increases, peaking at 4,000 in 2027. Region X remains consistent with annual peaks of around 2,000. Region XI predicts little to no cases across the years. Region XII shows a gradual decrease, peaking at 1,000 cases annually. Region XIII remains stable, with yearly peaks above 1,000. The region with the highest predicted cases is Region IV-A, with 12,500 predicted dengue cases. Overall, the highest number of dengue cases across all regions is consistently observed during the rainy seasons.

c. Forecasted Seasonal Patterns

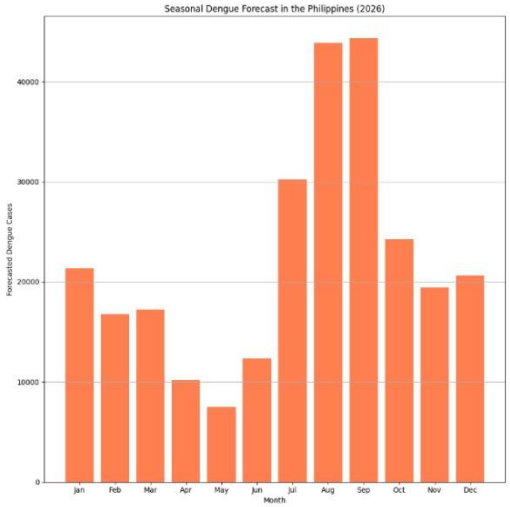


Fig.6 (Seasonal Dengue Forecast in the Philippines (2026))

The forecast for 2026 shows a clear seasonal pattern in dengue cases. From January to March, the number of cases is moderate. A noticeable decrease happens in April and May, with May having the lowest forecast of around 7,000 cases. Starting in June, cases begin to rise, reaching about 30,000 in July. The highest

numbers are expected in August and September, both with over 44,000 cases. From Octoberto December, the number of cases gradually decreases.

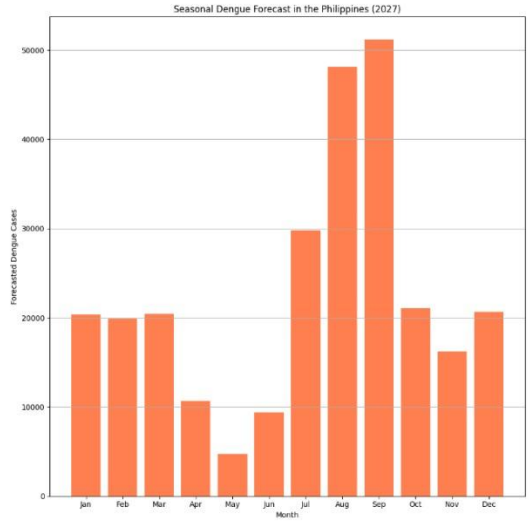


Fig.7 (Seasonal Dengue Forecast in the Philippines (2027))

Moving to 2027, the forecast follows a similar seasonal pattern. January to March have moderate cases. April to May was marked as the lowest, with May showing only around 5,000 cases, the lowest of the year. The trend changes in June, as it rises about 10,000. During July, a sharp increase happens around 30,000, then continuing to August with approximately 48,000, and September is the peak month with 50,000 cases. October shows a drop, then followed by November.

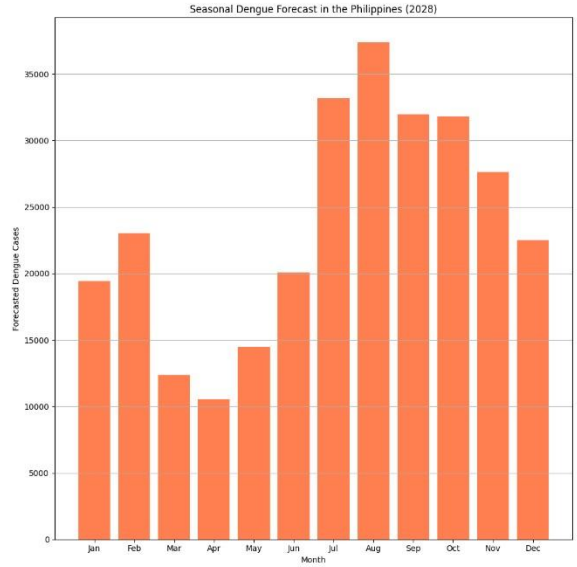


Fig.8 (Seasonal Dengue Forecast in the Philippines (2028))

Similarly, the 2028 forecast follows the same seasonal pattern. January starts with about 20,000 cases, and February sees a slight increase. March shows a drop, while May marks the beginning of a rise in cases. From June to October, dengue cases are high. June starts with 20,000, increasing to around 33,000–34,000 in July. August has the highest number, with over 36,000 cases. Although September and October show a slight decrease, the numbers remain high. A more apparent decrease begins at the end of the year.

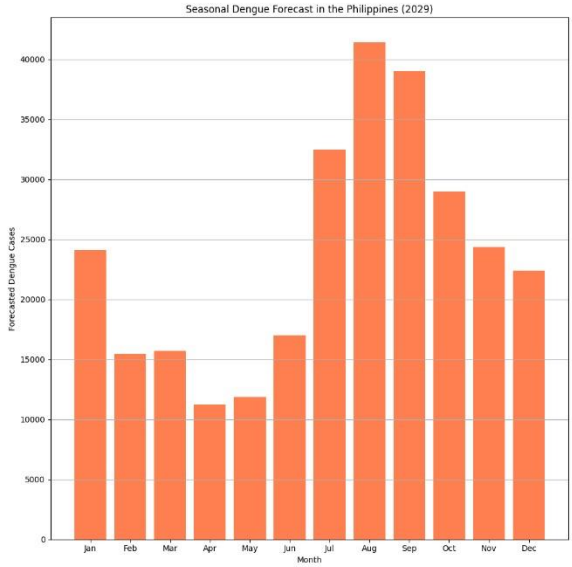


Fig.9 (Seasonal Dengue Forecast in the Philippines (2029))

For 2029, The start of the year, the forecast indicated moderate to high case numbers of dengue. During the second quarter of the year, April to June, dengue reaches its lowest level, with April recording the lowest forecasted cases. The peak dengue season spans from July to October, with the case numbers increasing dramatically. July shows a sharp jump from 17,000 cases in June to 32–33,000 cases. Followed by August, which recorded the highest monthly forecast for 2029, with over 41,000 cases. In the final months of the year, November and December, the data indicates a gradual decline in dengue activity.

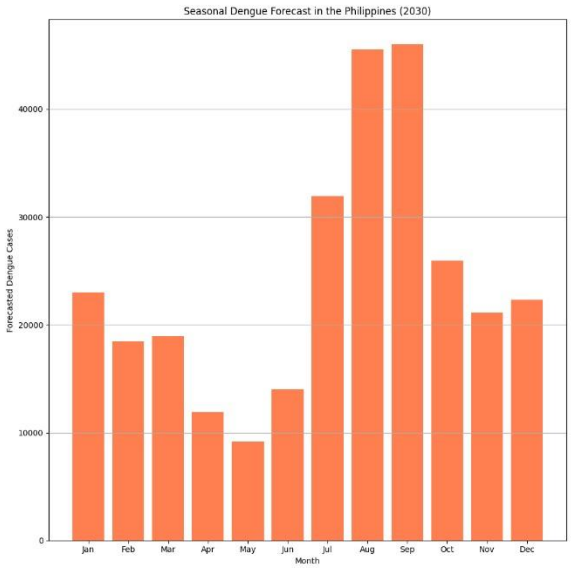


Fig.9 (Seasonal Dengue Forecast in the Philippines (2029))

Finally, the forecast for the year 2030, dengue cases in the Philippines for the year 2030 are forecasted to significantly increase at the beginning of July, with the highest number of cases in August and September, both exceeding 45,000 cases. The lowest number of cases occurs in May, with the cases dropping below 10,000.

Model performance

The cross-validation results of the Prophet model show how accurately the model can forecast dengue cases over different periods.

For short-term forecasts (30–90 days), the model performs very well, with a low Mean Absolute Percentage Error (MAPE) of around 1% or less. This means that the predicted dengue case counts are very close to the actual values in the short run.

As the forecast horizon increases (120–365 days), the model's accuracy declines. The MAPE increases to about 6%–9%, and RMSE values also rise, indicating larger errors in the predictions. This decline in accuracy is expected in longer-term forecasts due to the Increasing uncertainty In seasonal and external factors.

DISCUSSIONS

Descriptive Analysis of Historical Data

The analysis of dengue cases from years 2016 to 2020 shows clear patterns of how the disease spread across the Philippines. 2019 had the highest number of reported cases, with over 400,000 cases, while the lowest was in 2019, with fewer than 150,000. These changes may be related to differences in weather, mosquito control or public health action during those years.

In terms of seasonality, dengue cases follow a consistent pattern. The lowest numbers were recorded during the dry months of March to May, with April showing the fewest cases. Starting in June, cases began to rise, reaching their peak in July to September during the rainy season. A systematic review by Li et al. [18] showed that high ambient temperatures encourage vector spread and that a minimum temperature of 18 degrees Celsius is a critical limiting risk factor for dengue transmission. According to other international meta-analyses, the critical temperature range for dengue epidemics is between 22 and 29 degrees Celsius, and extremely high temperatures shorten the vector's survival time. This suggests that rainfall and humidity help mosquitoes breed, leading to more infections.

The data also shows differences between regions. Region IV-a had the highest total cases from 2016 to 2020, with around 160,000 cases, likely because of its large population and urban areas where mosquitoes thrive. Like earlier research, areas with a higher population density had a visually stronger temperature-dengueran association than areas with a lower population density [5]. In contrast, BARMM had the fewest cases, about 20,000, which could be due to fewer people, less urban development, or limited reporting. These results show that dengue risk varies across regions. Due to a lack of geographical and behavioural data, the study did not include additional modifying variables, such as forest cover or the presence of vegetation, open water containers, and the degree of intrahousehold crowding, which are known dengue risk factors. Unplanned urbanization encourages proximity to larval habitats, dengue-prone water storage practices, and rapid population movement [2]. As a result, dengue transmission may be increasing in rural areas.

A. Forecasted Trends

a. Forecast Results by Region

The forecasted dengue cases across the Philippine regions show a clear seasonal pattern, with most regions reaching their highest peak during the rainy season. These trends reflect the strong influence of climate on mosquito breeding and dengue transmission. Regions such as Region IV-A and NCR show high forecasted dengue cases, likely due to their dense populations, urban environments, and high levels of population movement. This aligns with the Department of Health's report identifying CALABARZON (10,759 cases), the National Capital Region (9,302 cases), and Central Luzon (8,652 cases) as the regions with the highest recorded dengue cases. [19] The consistency between historical data, forecasts, and current DOH statistics underscores the importance of regional and seasonal preparedness in dengue prevention and control.

b. Forecased Seasonal Patterns This seasonal dengue forecast aligns with the rainy season of the country, which typically starts in June. The sudden rapid rise of cases during June highlights the increase in dengue cases due to external conditions such as humidity and rainfall. From 2026 to 2030, the forecast shows a repeating pattern in dengue cases in the Philippines. The lowest case numbers usually happen in April and May, during the dry season when mosquitoes breed less. The highest numbers are seen from July to September, during the rainy season, when mosquito breeding increases. August often has the most cases. These trends show that dengue is affected by the weather and highlight the importance of mosquito control and prevention, especially before and during the rainy season. These seasonal findings align with WHO findings, which link dengue transmission to environmental factors such as rainfall, humidity and temperature, which all are favourable conditions for mosquito breeding and viral replication. [5]

c. Model performance

The cross-validation results of the Prophet model show how accurately the model can forecast dengue cases over different periods.

For short-term forecasts (30–90 days), the model performs very well, with a low Mean Absolute Percentage Error (MAPE) of around 1% or less. This means that the predicted dengue case counts are very close to the actual values in the short run.

As the forecast horizon increases (120–365 days), the model's accuracy declines. The MAPE increases to about 6%–9%, and RMSE values also rise, indicating larger errors in the predictions. This decline in accuracy is expected in longer-term forecasts due to the increasing uncertainty in seasonal and external factors.

CONCLUSION AND RECOMMENDATION

This study identified clear temporal and regional patterns in dengue incidence across the Philippines from 2016 to 2020 and provided forecasts for 2026 to 2030 using the Prophet model. Results revealed strong seasonality in dengue trends, with cases consistently peaking during the rainy months of July to September and dropping during the dry season. Region IV-A recorded the highest historical and forecasted case counts, while BARMM consistently showed the lowest. National-level forecasts suggest recurring peaks each year, particularly in August and September, with 2019 marking the highest recorded historical total of over 400,000 cases. These findings demonstrate the critical role of seasonal and environmental factors in shaping dengue outbreaks across regions.

While the Prophet model demonstrated strong performance in short-term forecasts, its accuracy decreased in long-term projections, particularly beyond 120 days. This suggests the need for continuous model retraining and incorporation of more up-to-date and exogenous variables (e.g., rainfall, temperature, mobility data) to improve future predictions.

Given these patterns, it is recommended that public health authorities strengthen early warning systems and prioritize intervention efforts during highrisk periods and in regions with consistently high case projections. Strategies may include intensified vector control, public awareness campaigns, and health resource allocation, particularly in Region IV-A and during mid-year months. The integration of regional forecasting into national planning can enhance preparedness and response efforts.

Further studies incorporating more recent and diverse data sources, including climatic, demographic, and socioeconomic factors, are encouraged to refine predictions and support more targeted and effective dengue control programs.

REFERENCES

[1]World Health Organization, "Dengue and severe dengue," *World Health Organization*, Apr. 23, 2024. https://www.who.int/news-room/factsheets/detail/dengue-and-severe-dengue

[2]eClinicalMedicine, "Dengue as a growing global health concern," *eClinicalMedicine*, vol. 77, p. 102975, Nov. 2024, doi: https://doi.org/10.1016/j.eclinm.2024.102975.

[3]WHO, "Dengue," *www.who.int*. https://www.who.int/southeastasia/healthtopics/dengue-and-severe-dengue

[4] L. Bravo, V. G. Roque, J. Brett, R. Dizon, and M. L'Azou, "Epidemiology of Dengue Disease in the Philippines (2000–2011): A Systematic Literature Review," *PloS Neglected Tropical Diseases*, vol. 8, no. 11, p. e3027, Nov. 2014, doi: https://doi.org/10.1371/journal.pntd.0003027.

[5] J. Snyder and G. Maglasang, "Dengue in Cebu City, Philippines: A Pilot Study of Predictive Models and Visualizations for Public Health," *The American Journal of Tropical Medicine and Hygiene*, vol. 110, no. 1, pp. 179–187, Jan. 2024, doi: https://doi.org/10.4269/ajtmh.23-0250.

[6] R. May-os, "DENGUE CASES IN THE PHILIPPINES 2024," *ResearchGate*, Sep. 08, 2024. https://www.researchgate.net/publication/383860548_DENGUE_CASES_IN_THE_PHILIPPINES_2024

[7]Xerxes Seposo, S. Valenzuela, and Geminn Louis Apostol, "Socio-economic factors and its influence on the association between temperature and dengue incidence in 61 Provinces of the Philippines, 2010–2019," *PLOS Neglected Tropical Diseases*, vol. 17, no. 10, pp. e0011700–e0011700, Oct. 2023, doi: https://doi.org/10.1371/journal.pntd.0011700.

[8]World Health Organization, "Dengue and severe dengue," *World Health Organization*, Apr. 23, 2024. https://www.who.int/news-room/factsheets/detail/dengue-and-severe-dengue

[9]Xerxes Seposo, S. Valenzuela, and Geminn Louis Apostol, "Socio-economic factors and its influence on the association between temperature and dengue incidence in 61 Provinces of the Philippines, 2010–2019," *PLOS Neglected Tropical Diseases*, vol. 17, no. 10, pp. e0011700–e0011700, Oct. 2023, doi: https://doi.org/10.1371/journal.pntd.0011700.

[10] S. Sharmin, K. Glass, E. Viennet, and D. Harley, "Interaction of Mean Temperature and Daily Fluctuation Influences Dengue Incidence in Dhaka, Bangladesh," *PLOS Neglected Tropical Diseases*, vol. 9, no. 7, p. e0003901, Jul. 2015, doi:https://doi.org/10.1371/journal.pntd.0003901.

[11] R. Nuranisa, Y. B. Maryanto, and M. A. Isfandiari, "CORRELATION OF FREE LARVAE INDEX AND POPULATION DENSITY WITH DENGUE FEVER

INCIDENCE RATE," *The Indonesian Journal of Public Health*, vol. 17, no. 3, pp. 477–487, Nov. 2022, doi: https://doi.org/10.20473/ijph.v17i3.2022.477487.

[12] S. Bhatt *et al.*, "The global distribution and burden of dengue," *Nature*, vol. 496, no. 7446, pp. 504–507, Apr. 2013, doi: https://doi.org/10.1038/nature12060.

[13] M. F. Vincenti-Gonzalez *et al.*, "Spatial Analysis of Dengue Seroprevalence and Modeling of Transmission Risk Factors in a Dengue Hyperendemic City of Venezuela," *PLOS Neglected Tropical Diseases*, vol. 11, no. 1, p. e0005317, Jan. 2017, doi: https://doi.org/10.1371/journal.pntd.0005317.

[14] L. Zheng, H. Ren, R. Shi, and L. Lu, "Spatiotemporal characteristics and primary influencing factors of typical dengue fever epidemics in China," vol. 8, no. 1, Mar. 2019, doi: https://doi.org/10.1186/s40249-019-0533-9.

[15] S. Polwiang, "The time series seasonal patterns of dengue fever and associated weather variables in Bangkok (2003-2017)," *BMC Infectious Diseases*, vol. 20, no. 1, Mar. 2020, doi: https://doi.org/10.1186/s12879-020-4902-6.

[16] K. Mulligan, J. Dixon, C.-L. Joanna Sinn, and S. J. Elliott, "Is dengue a disease of poverty? A systematic review," *Pathogens and Global Health*, vol. 109, no. 1, pp. 10–18, Dec. 2014, doi: https://doi.org/10.1179/2047773214y.0000000168.

[17] R. Nuranisa, Y. B. Maryanto, and M. A. Isfandiari, "CORRELATION OF FREE LARVAE INDEX AND POPULATION DENSITY WITH DENGUE FEVER
INCIDENCE RATE," *The Indonesian Journal of Public Health*, vol. 17, no. 3, pp. 477–487, Nov. 2022, doi: https://doi.org/10.20473/ijph.v17i3.2022.477487.

[18] Y. Li, Q. Dou, Y. Lu, H. Xiang, X. Yu, and S. Liu, "Effects of ambient temperature and precipitation on the risk of dengue fever: A systematic review and updated meta-analysis," *Environmental Research*, vol. 191, p. 110043, Dec. 2020, doi: https://doi.org/10.1016/j.envres.2020.110043.

[19]"DO NOT BE COMPLACENT ABOUT THE DENGUE THREAT FROM Aedes aegypti —DOH – Department of Health," Department of Health, Mar. 07, 2025. https://doh.gov.ph/presshttps://doh.gov.ph/press-release/do-not-be-complacent-about-the-dengue-threat-from-aedes-aegypti-doh/release/do-not-be-complacent-about-the-denguehttps://doh.gov.ph/press-release/do-not-be-complacent-about-the-dengue-threat-from-aedes-aegypti-doh/threat-from-aedes-aegypti-doh/

[20] Jhon Vincent Gupo, "Philippines Dengue Cases 2016-2020," *Kaggle.com*, 2016. https://www.kaggle.com/datasets/vincentgupo/dengue-cases-in-the-philippines/code (accessed Jul. 01, 2025).